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Introduction to: EDA/GSR

EDA (also commonly referred to as GSR) measures emotional arousal via levels of perspiration on the surface of the skin, indicating the intensity of emotional response to stimuli.

Background

- When we are exposed to images, videos, events, or other kinds of emotionally loaded stimuli, the electrical conductivity of our skin subtly changes.
- Skin conductivity is controlled on an entirely subconscious level.
- In other words, we cannot consciously control our skin conductance, and so GSR is the perfect marker for emotional arousal, as it offers clear insights into the physiological and psychological processes of an individual.
- This metric is also referred to as Electrodermal Activity (EDA) or Skin Conductance (SC).

How it Works

- When sweat glands are triggered, they secrete moisture through pores towards the skin surface. By changing the balance of positive and negative ions in the secreted fluid, electrical current flows more readily, resulting in measurable changes in skin conductance.
- Whenever we are emotionally aroused, GSR data shows distinctive patterns that are visible with bare eyes and that can be quantified statistically.

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- When experiencing emotional arousal, the level of sweat on our hands can be noticeable if we are highly excited or nervous. However, most of the time, the emotions we experience lead to minimal changes in the sweat level that can only be captured by the GSR signal.
- GSR response can also be affected by other factors such as familiarity and prior associations with the stimulus.

Applications

"*By 1972, more than 1,500 articles on GSR had been published in professional magazines, and nowadays GSR is regarded as the most popular method for investigating human psychophysiological phenomena.*" - Boucsein (2014)

- With GSR, the impact of any emotionally arousing content, product or service can be tested. Physical objects, videos, images, sounds, odors, food and other sensory stimuli as well as thought experiments and mental images can be analyzed.
- GSR is prevalent in psychological research, clinical research and psychotherapy, consumer neuroscience and marketing, media and ad testing, usability testing and UX design.

Quantification

- A "GSR peak" is a biological indicator that something relevant happened at that moment – an emotional connection was made.
- One clear way to visualize the GSR peaks is by calculating GSR peaks per minute. GSR peaks per minute indicates, on average, how many emotional events occurred every 60 seconds. The higher the number, the

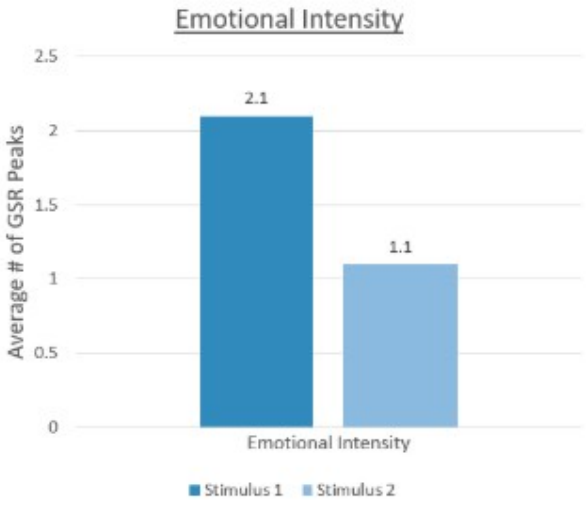
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more of an emotional response the viewers were having during the content or experience.

- Please refer to our Help Center articles on GSR data analysis and export on iMotions.

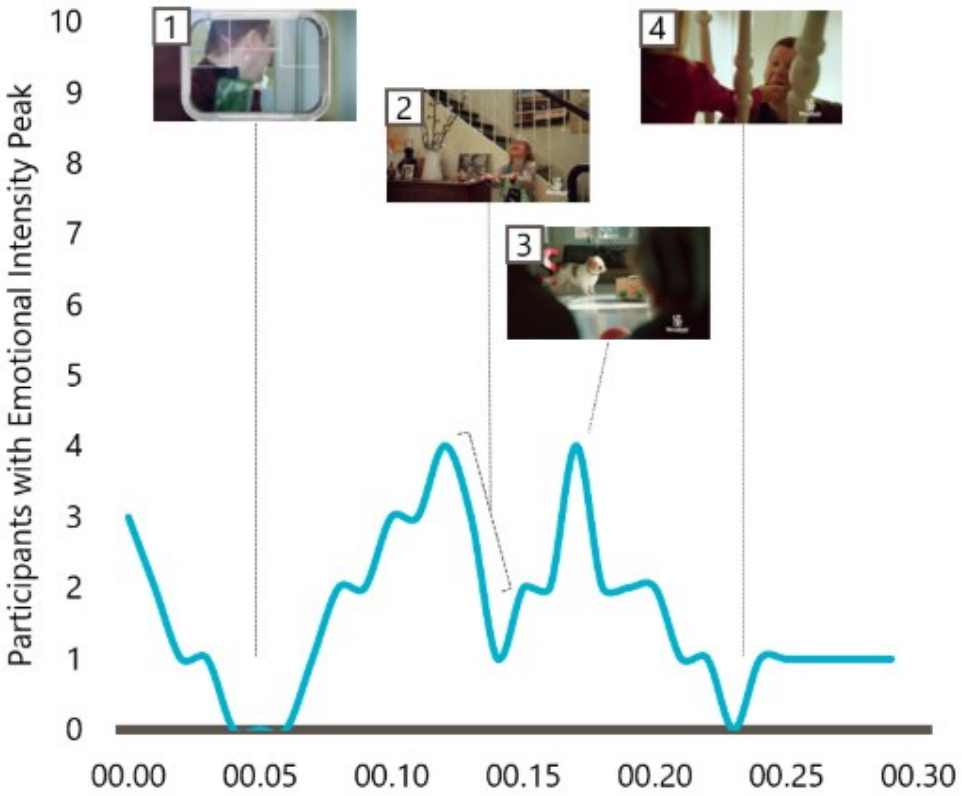
Data Visualizations

- Overall scores can be compared, like in the example below, to indicate which of the stimuli tested evoke more of an emotional response in participants or viewers. In the example, Stimulus 1 creates one additional “emotional event” every 60 seconds relative to Stimulus 2, indicating that it likely makes a stronger emotional connection with viewers.



- In addition to overall scores, emotional intensity can be measured on a moment to moment (M2M) basis to better understand the participants’ emotional journey while being exposed to video media content. This method can be used to identify key moments of the content that are connecting, as well as other portions that may not be engaging to viewers. M2M traces are most effective when combined with eye tracking and facial expression analysis, which can provide information on how the media directs audience gaze and what emotions they expressed during the content. For example, eye tracking may reveal that viewers completely miss key elements on the screen in too-dark scenes, while FEA may show that only a portion of the audience smiled after a key joke was made. This multimodal approach allows researchers to investigate avenues to optimize their content.

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What GSR Does Not Measure

- It is important to note that while GSR can measure if there is an emotional response and the intensity of an emotional response, it does not measure the valence, i.e. whether that emotional response is positive or negative. For this reason, it is best to combine GSR with other biometric measures such as eye tracking, EEG, and FEA to provide additional insight into why an individual or group of individuals may exhibit a particular strength of emotional response to certain stimuli and experiences.

Academic Applications

Historically, GSR is an incredibly widespread method for testing how humans respond emotionally towards various stimuli. In psychological research, experimenters may use GSR to investigate how responses are affected by stimulus properties, personality characteristics, social expectancies, and interaction of cultural aspects and individual learning histories. Clinically, GSR can be used during cognitive-behavioral therapy to provide a quantitative measure of physiological arousal of the patient and thus the severity of the disease of interest and the success of therapeutic intervention.

Commercial Applications

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GSR is an incredibly useful modality to add to commercial research, intersecting many different fields of interest. In consumer neuroscience and marketing, GSR can be measured to assess emotional arousal towards products and decisions made primarily based on affective and subconscious processes. These findings can be utilized to enhance products, assess market segments, or identify target audiences and buyer personas. In media and ad testing, TV ads, trailers, and full length shows or films can be shown to individual participants or focus groups to monitor emotional engagement and identify key frames or scenes that “just don’t work.” GSR is also heavily utilized in usability testing and UX design to provide insights into the stress level of users during interaction with new website content, user interfaces, and online forms to illuminate opportunities for optimization.

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